

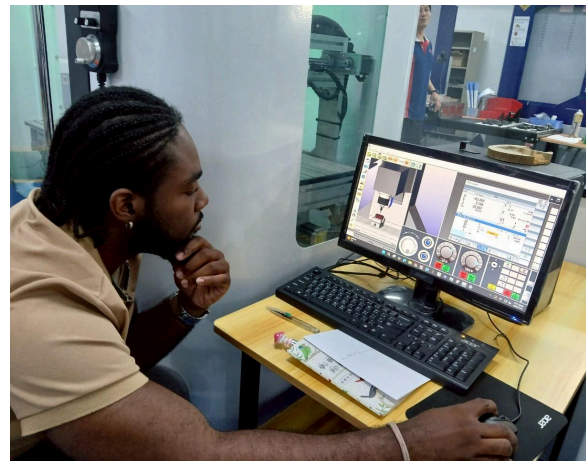
Progress Report

My goal is to develop a swarm intelligence framework to coordinate two hexapod robots in an indoor warehouse environment, enabling efficient collaboration and task execution. This includes implementing a robust indoor positioning system to ensure accurate navigation across uneven and cluttered terrain. I have set up the perimeter for four beacons and tested the Link-PG app to track the tag within the designated space. However, due to metal and magnetic interference, there is currently a negligible error of approximately 10 cm. I am actively working to minimize this error for more precise tag detection within the perimeter.

Through my participation in the Development Application Base of Engineering Talents (iDABET), I have gained valuable technical skills and insights, especially regarding the semiconductor industry, Taiwan's largest and most influential sector. This experience has not only broadened my knowledge but also helped guide the direction of my thesis. One of the courses, CNC milling simulation, inspired me to create a simulation of my own using Unity Hub for research purposes.



Perimeter Set-up of beacons



CNC Simulation Course